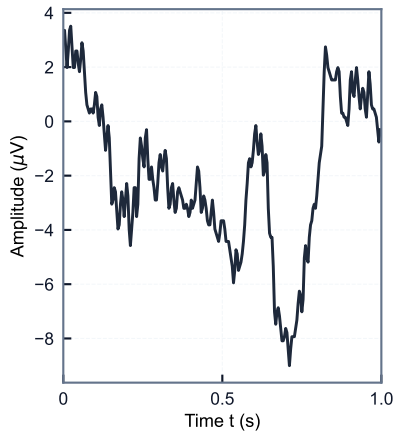
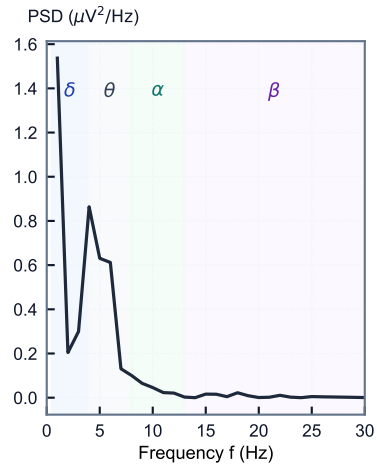


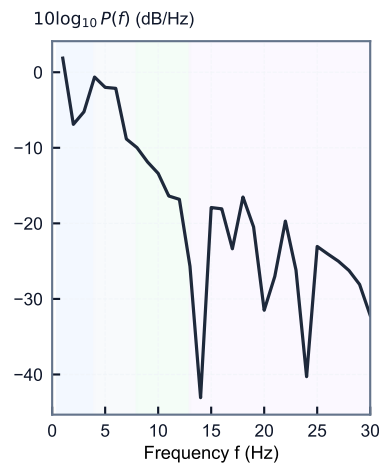
(a) Raw EEG Epoch $x(t)$



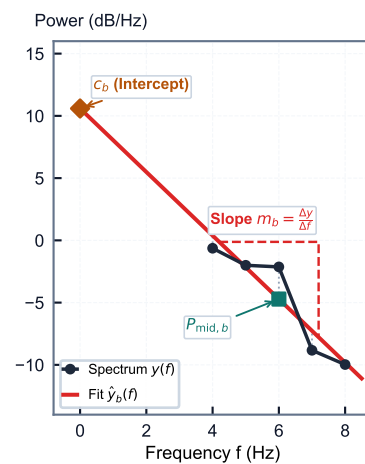
(b) Welch PSD $P(f)$



(c) Log-PSD Dynamics $y(f)$



(d) Sub-Band OLS Fit $\hat{y}_b(f)$



(e) Biomarker Vector $\mathbf{x}_{slope}$

1. Sub-Band Vector:
 $\mathbf{f}_b = [m_b, c_b, P_{mid,b}]^T$
2. Channel Vector:
 $\mathbf{f}_c = [\mathbf{f}_\delta^T, \mathbf{f}_\theta^T, \mathbf{f}_\alpha^T, \mathbf{f}_\beta^T]^T$
3. Channel Dim:
 $\dim(\mathbf{f}_c) = 4 \times 3 = 12D$
4. Multichannel:
 $\mathbf{x}_{slope} = [\mathbf{f}_1^T, \dots, \mathbf{f}_C^T]^T$

$$\mathbf{x}_{slope} \in \mathbb{R}^{C \times 12}$$

($C = 18 \rightarrow 216D$)